AI ACADEMY · PYTHON READINESS BRIDGE · STUDENT LESSON

Boolean conditions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

Name each requirement. Test the boundary. Combine deliberately.

Translate a written condition into separate Boolean checks without replacing AND with OR.

Keep one habit: Minutes/quiet trace → inclusive age interval → five activity cases → distinguishing mutations → independent requirements.

MAIN LESSON · SECTION 1

Must both requirements hold, or is either enough?

> Translate the written condition and test the exact boundary.

Your learning goals

I can translate stated requirements into **comparison**s.

I can combine **Boolean** values with and or and not.

I can choose a case that distinguishes two conditions.

Recall before notes

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.
- Without opening the old lesson, explain one key idea from Expressions and operator order. Give an example and a mistake that the idea helps you avoid.
- Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.
- After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.
- This week you will practise:
- Translate stated requirements into **comparison**s
- Combine **Boolean** values with and or and not
- Choose a case that distinguishes two conditions
- What type is False and how does it differ from "False"?

> **My starting explanation**

Your end goal

Trace **boolean** conditions and explain the changed program before running it.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

At most twelve minutes AND quiet

> A time-only pass is different from full suitability.

Read the complete activity requirements

A fictional activity must take at most **12 minutes** and be quiet. An activity taking exactly **12 minutes** meets the time requirement. A quiet 15-minute activity does not.

Before writing code, underline each requirement and decide whether all requirements must hold or whether alternatives are allowed.

State input types and comparison syntax

Use a fresh conditions.py file with nonnegative integer minutes and an actual **Boolean** quiet flag. A **comparison** produces True or False for these values. The expression minutes <= 12 includes 12; minutes < 12 excludes it.

Use == for equality and != for inequality. A single = assigns a value to a name; it does not ask whether two values are equal.

Your first explanation

List the given facts and one fact that is not supplied.

MAIN LESSON · SECTION 3

Calculate each Boolean before combining them

The supplied inputs are a nonnegative integer and a real Boolean flag; no type validator is implemented.

Trace separate and combined conditions

Two requirements must both hold

INPUT CASE	TIME CHECK	COMBINED RESULT
12 min, quiet False	12 <= 12 is True	True and False is False
12 min, quiet True	12 <= 12 is True	True and True is True
15 min, quiet True	15 <= 12 is False	False and True is False

Passing only one requirement does not make the activity suitable.

Complete activity-condition script

```
minutes = 12
quiet = False
time_ok = minutes <= 12
suitable = time_ok and quiet
print(time_ok)
print(suitable)
```

Explain time success but suitability failure

The script prints True then False. The first result concerns time only. The second requires time_ok and quiet. These are separate claims.

Trace each **input** before explaining the combined result; a result without its conditions can be misleading.

Explain the deciding check

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare an included boundary with a missing requirement

> Changing <= to < or AND to OR changes the written policy.

Predict the supported change

- For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time **comparison** and final suitability values before running the program.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured variant

- Change: Change age to 16.

▼ **Check after predicting**

Prediction: The result becomes False.

Reason: The lower condition remains True, but the upper condition fails.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace both ends of a separate inclusive interval

> One boundary check alone cannot enforce the full interval.

Read the complete toy age contract

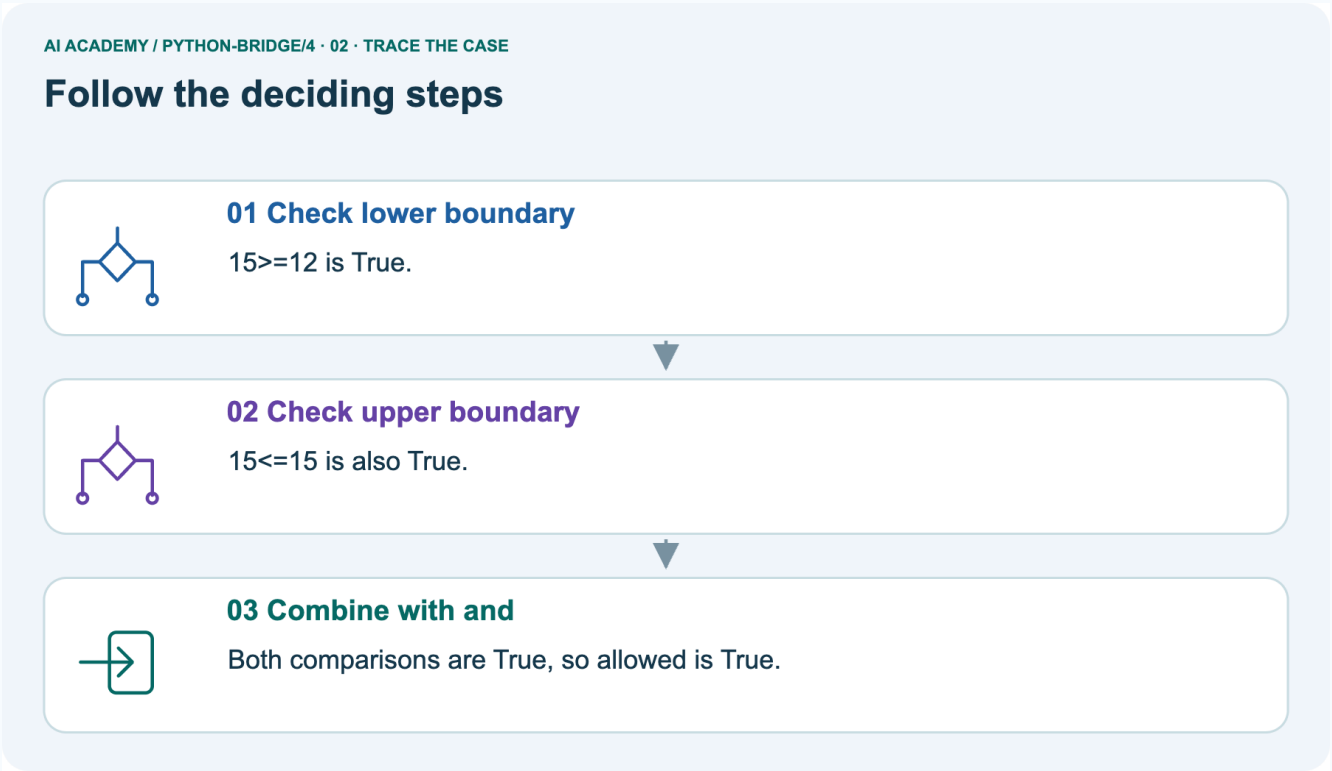
Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

This toy club exercise uses only supplied integer ages, not actual pupils' information. It checks a numeric interval, not identity, permission or arbitrary input validity. At age 11 the lower check rejects; at 16 the upper check rejects. Both boundaries 12 and 15 are included. The activity example's minutes/quiet fields are a different contract.

Truth-table statements here use Boolean operands; Python's more general and/or operand behavior is outside this exercise.

Trace both comparisons and AND



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check lower boundary	15>=12 is True.
2. Check upper boundary	15<=15 is also True.
3. Combine with and	Both comparisons are True, so allowed is True.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **Boolean** condition represents a yes-or-no result. And requires both conditions; checking only one boundary would accept values outside the intended interval.

Complete three-value comparison program

```
age = 15
lower_ok = age >= 12
upper_ok = age <= 15
allowed = lower_ok and upper_ok
print(lower_ok)
print(upper_ok)
print(allowed)
```

Inspect below/at/above boundary cases

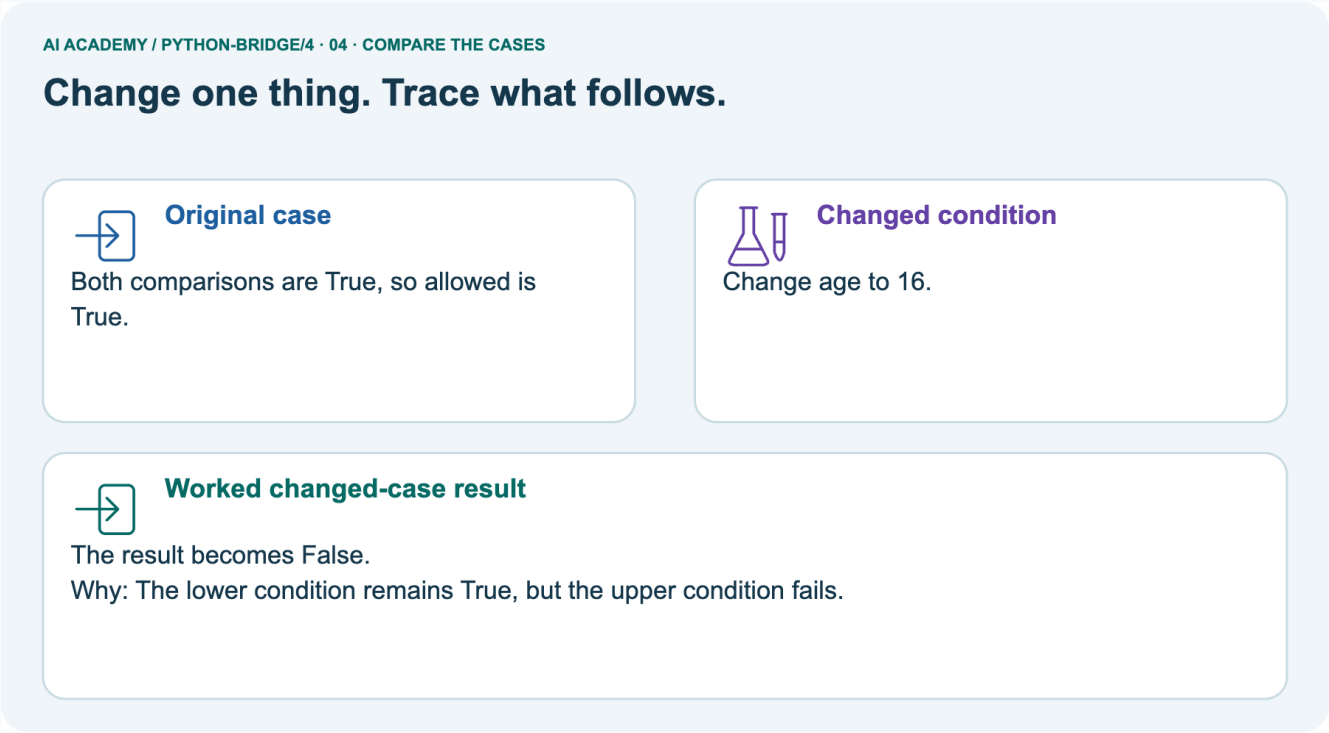
Supplied integer age	age>=12	age<=15	Both requirements
11	False	True	False

Supplied integer age	age>=12	age<=15	Both requirements
12	True	True	True
15	True	True	True
16	True	False	False

Predict changing 15 to 16

- Change one thing: Change age to 16. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

State the validation/authority limit

- This condition checks a declared numeric interval, not identity or authorisation; unexpected **input** types need separate handling.

- Read the labelled picture: Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

▼ Reference · Boolean AND/OR/NOT truth table

A	B	A and B	A or B
False	False	False	False
False	True	False	True
True	False	False	True
True	True	True	True

With **Boolean** operands, and needs both parts True. **Inclusive or** needs at least one True, so both True is allowed. not reverses a **Boolean**: not True is False and not False is True.

Python can also apply and and or to other types; keep operands **Boolean** here so this table directly describes the result.

> **A game requires a pencil or a crayon. If both are available, does inclusive or accept the case? Explain how this differs from requiring exactly one.**

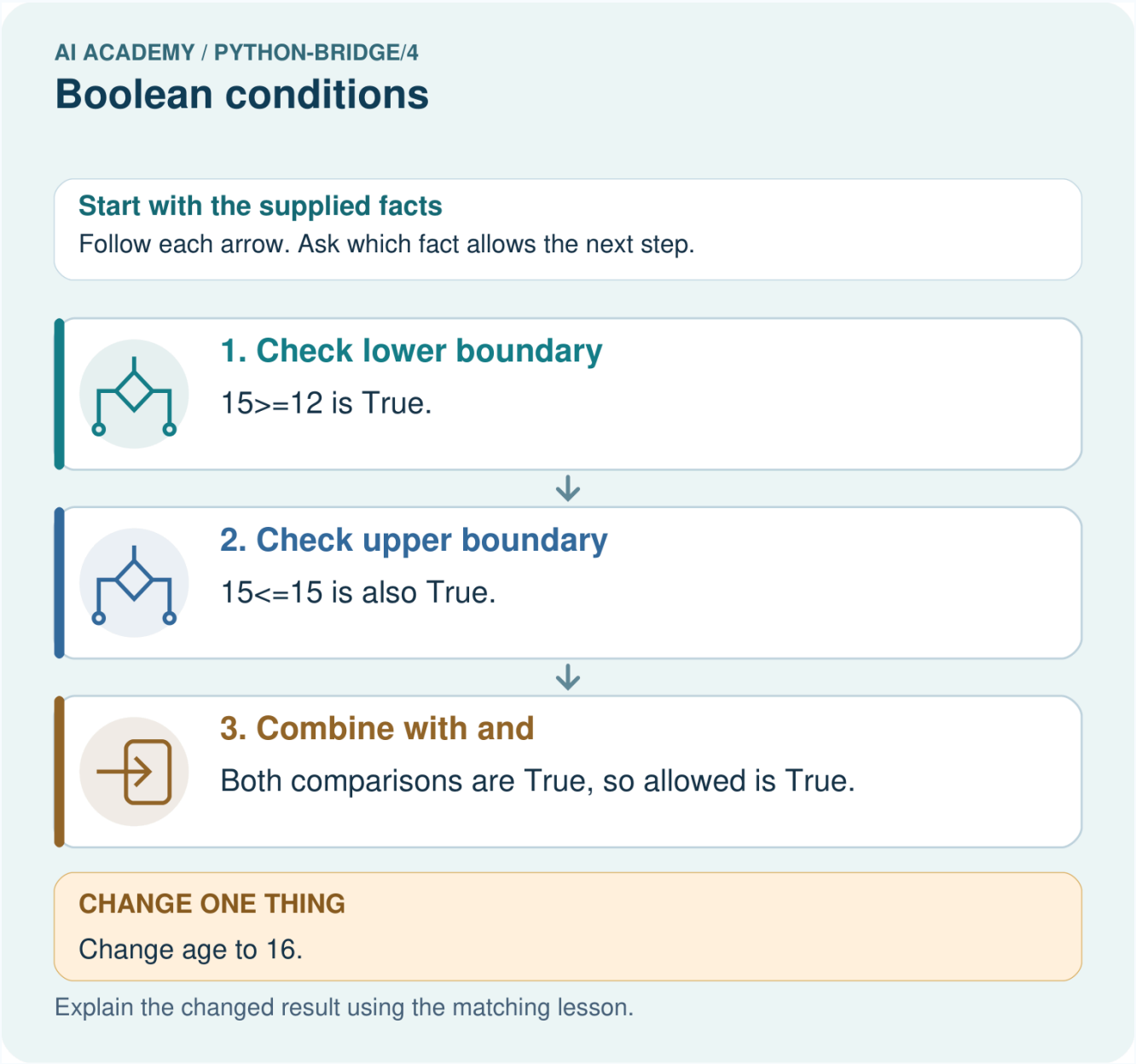
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Boolean conditions

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Write both comparisons in an OR expression

> A bare second value is not a repeated equality check.

Run the separate range/NOT/equality script

```
minutes = 12
quiet = True
print((minutes >= 0) and (minutes <= 12))
print(not quiet)
print(minutes == 12)
```

Compare correct and incomplete alternative checks

This prints True, False and True. Parentheses make the two range checks clear. To express minutes equal to 8 or 12, write (minutes == 8) or (minutes == 12).

Do not write minutes == 8 or 12: the number 12 is not a second equality **comparison**. Later lessons examine truth testing of other types.

Keep supplied-input and permission limits

The rule uses values supplied by a person. It does not verify that the activity is really quiet, and a True result grants no permission to act.

Our **input** contract already assumes a valid integer and **Boolean**; handling wrong types is a separate task.

Explain inclusive OR for pencil/crayon

- Complete all four **Boolean** combinations for pencil or crayon. Explain the both-available case and the effect of not pencil.

State one repair and the checks you would rerun.

▼ **Earlier wording, alternatives and picture transcripts**

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the minutes-and-quiet decision with less help

> Record time and full suitability separately.

Explain each condition

- Write a condition for an activity taking at most **12 minutes** and being quiet. Explain why exactly **12 minutes** is included.

A game requires a pencil or a crayon. If both are available, does inclusive or accept the case? Explain how this differs from requiring exactly one.

Explain your prediction to a partner before comparing results.

Complete the W2 cases

Use workbook W2 for the connected example “**Boolean** conditions”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Check the at-most boundary

- Use the worked case for **Boolean** conditions. Proposed explanation: “At most excludes the boundary”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain both age boundaries

- Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

My explanation using a pictured step and the given facts

Reserve independent requirements

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

Practise with less help: Trace separate checks

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Test five complete activity cases and two rule mutations

> Choose a counterexample that distinguishes the intended rule from a tempting replacement.

All supplied inputs and separate predictions

Complete supplied activity case	time_ok	suitable with AND	What the case reveals
11 minutes, quiet=True	True	True	Ordinary eligible case
12 minutes, quiet=False	True	False	Time pass is not full suitability; OR would wrongly accept
13 minutes, quiet=True	False	False	Quiet alone cannot overcome the time limit
12 minutes, quiet=True	True	True	<= includes the exact boundary; < would reject

Complete supplied activity case	time_ok	suitable with AND	What the case reveals
0 minutes, quiet=False	True	False	Valid supplied zero time still needs quiet

Predict, change a rule and preserve the mismatch

1. Use the complete activity script and edit only the supplied minutes/quiet inputs for each row. Predict time_ok and suitable before rerunning the whole file. Compare one intentional rule change at a time: <= to < at exactly 12, or AND to OR when only one condition holds. Preserve both original and changed results.

   Use the truth table to explain inclusive OR when both Boolean conditions are true; it does not mean exactly one. Keep input-validation questions separate: these cases already supply nonnegative integer minutes and actual booleans. Wrong types and physical quietness are not checked by this program. Keep the independent genre/pages/availability script for the individual attempt.

Complete original activity and range scripts

```
minutes = 12
quiet = False
time_ok = minutes <= 12
suitable = time_ok and quiet
print(time_ok)
print(suitable)
```

```
minutes = 12
quiet = True
print((minutes >= 0) and (minutes <= 12))
print(not quiet)
print(minutes == 12)
```

Keep the evidence record

For (minutes, quiet) cases (11, True), (12, False), and (13, True), write the time **comparison** and final suitability values before running the program.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: preserve all requirements in the selected script

> An OR result can be true while a mandatory genre condition is false.

Selected complete task and scope

Predict all four **boolean**s. Eligibility requires science AND at most100 pages AND availability. Repeat with genre="history" and all other values unchanged. Which OR result could wrongly suggest the genre requirement passed?

```
pages = 90
genre = "science"
available = False
print(genre == "science" and pages <= 100)
print((genre == "science" and pages <= 100) and available)
print(not available)
print(genre == "science" or pages <= 100)
```

Use only the supplied science/page/availability requirements and values. Recompute each printed expression in order when changing the genre; leave the other inputs unchanged. A partial requirement check must not be reported as complete eligibility, and the stated flags are supplied data rather than verified real-world availability.

Your response and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Additional practice · Earlier Boolean transfer

A fictional puzzle qualifies when its duration is at least 5 and at most 9 minutes and its available flag is True. Write the condition. Predict results for (5, True), (9, True), (10, True), and (7, False).

Choose a case that detects replacing and with or between the range check and available.

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Combine requirements without replacing AND

Attempt this without hints

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the written requirement beside the Boolean code

> A True result should state exactly which checks passed.

This week’s contribution

Add a labelled **boolean** conditions **evidence** sheet to Python readiness portfolio. Show how you can translate stated requirements into **comparison**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project checkpoint and evidence

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain each input, boundary and connector

> A correct Boolean calculation does not verify its source facts or grant authority.

Explain without notes

- Close your notes. Explain this goal in your own words: Translate stated requirements into **comparison**s

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · The words used in this procedure**

REFERENCE · SECTION 5

Reference · The words used in this procedure

> Explain each term using a particular code line.

Vocabulary and response

Word	Meaning
Comparison	An expression that checks a relation between values
Boolean	A type with the values True and False
Conjunction	A requirement that both conditions hold
Inclusive or	A condition satisfied when either or both parts hold
Counterexample	A case that shows a claim or rule is wrong

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain, apply and challenge

REFERENCE · SECTION 10

Extra challenge · Explain, apply and challenge

> Choose the next useful task after supported practice.

Choose a challenge

- Write a condition for an activity taking at most **12 minutes** and being quiet. Explain why exactly **12 minutes** is included.
- Complete all four **Boolean** combinations for pencil or crayon. Explain the both-available case and the effect of not pencil.
- Explain why a True suitability result does not prove that the supplied duration is accurate or that a child has permission to start the activity.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Earlier wording, alternatives and picture transcripts

Explain → apply → challenge

▼ Reference · Inspect your evidence

REFERENCE · SECTION 12

Reference · Inspect your evidence

> Use the record after practice rather than revealing the independent answer.

Self-check and explanation

- Testing only a quiet 8-minute activity cannot distinguish and from or because both pass. Use a case where exactly one requirement holds.

  Keep expected results beside observed results and explain which requirement caused failure. Next week a program will choose an action using a condition.

- Explain why a True suitability result does not prove that the supplied duration is accurate or that a child has permission to start the activity.

> **For the original at-most-12-minutes and quiet rule, choose one input that reveals changing <= to < and a different input that reveals changing and to or. State the expected correct result for each.**

My limited claim and a useful next check

▼ **Extra practice · A complete trace**

REFERENCE · SECTION 13

Extra practice · A complete trace

> Predict before opening the supplied trace.

Explain the input and procedure

- Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Check the complete trace

Two requirements must both hold

INPUT CASE	TIME CHECK	COMBINED RESULT
12 min, quiet False	12 <= 12 is True	True and False is False
12 min, quiet True	12 <= 12 is True	True and True is True
15 min, quiet True	15 <= 12 is False	False and True is False

Passing only one requirement does not make the activity suitable.

▼ Earlier wording, alternatives and picture transcripts

Complete evidence record

▼ Reference · Earlier route checklist

REFERENCE · SECTION 15

Reference · Earlier route checklist

> Keep one main lesson route and use this as a reminder.

Checklist and first prediction

By the end, I can	How I will show it
Translate stated requirements into comparisons	Explain the deciding step, show a trace or test, and state one limit.
Combine Boolean values with and or and not	Explain the deciding step, show a trace or test, and state one limit.
Choose a case that distinguishes two conditions	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 17

Extra practice · Predict, check and repair

> Use only after an attempt; keep the reserved assessment separate.

First question and hint

- Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.

- Which labelled step matches this detail? Both **comparison**s are True, so allowed is True.
- 1. Check lower boundary
- 2. Check upper boundary
- 3. Combine with and
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Combine with and Both **comparison**s are True, so allowed is True. A **Boolean** condition represents a yes-or-no result.

And requires both conditions; checking only one boundary would accept values outside the intended interval.

Changed-case question and hint

- Changed-case practice: Change age to 16. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. The result becomes False.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed explanation

After your attempt, compare with the supported explanation: The result becomes False. The lower condition remains True, but the upper condition fails.

Repair and keep the record

Review Expressions and operator order (python-bridge/3). Goal: Evaluate numeric expressions using Python's grouping rules. Short practice: Trace the operation order in both expressions.

Identify which picture step changes with parentheses and predict both results when the multiplier becomes four.. Return to this week and explain what you changed.

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name.

Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · The six-picture journey

REFERENCE · SECTION 18

Reference · The six-picture journey

> Choose the picture needed for the present question.

01. Boolean conditions: the case

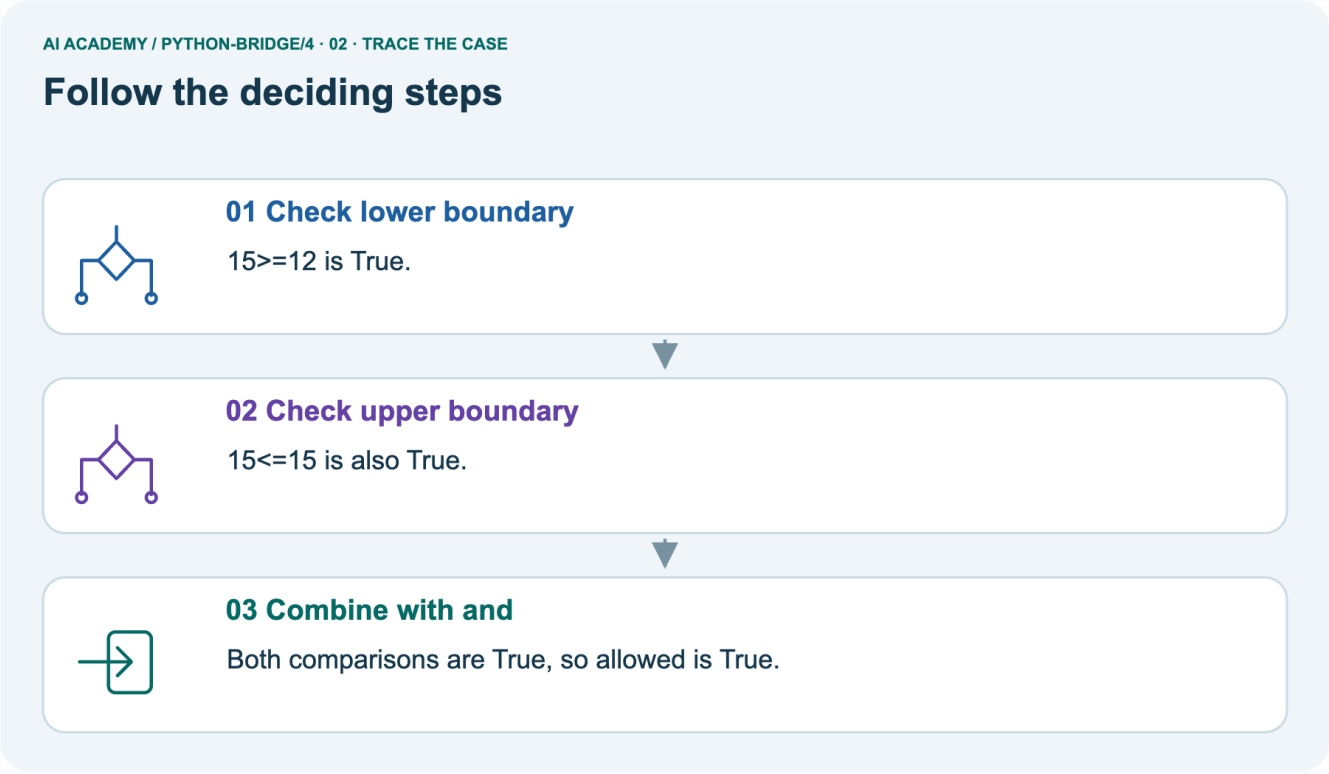


Follow the labelled picture with the matching explanation. Open the vector diagram

Two conditions for entry. A toy club accepts integer ages from 12 through 15 inclusive.
Supplied facts: Python 3: age=15; allowed=(age>=12) and (age<=15); print(allowed). No type validation is implemented here.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

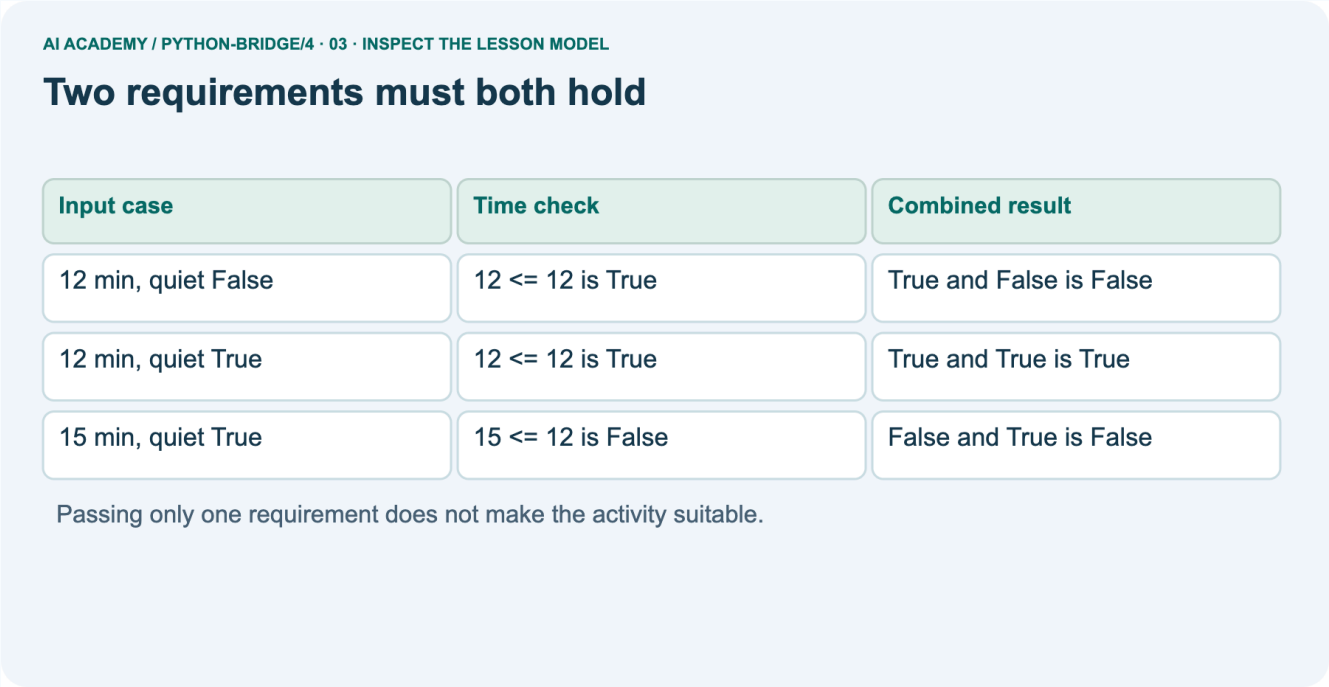


Follow the labelled picture with the matching explanation. Open the vector diagram

Check lower boundary: 15>=12 is True. → Check upper boundary: 15<=15 is also True. → Combine with and: Both **comparison**s are True, so allowed is True.

Point to each numbered stage and explain which recorded **input** it uses.

03. Two requirements must both hold

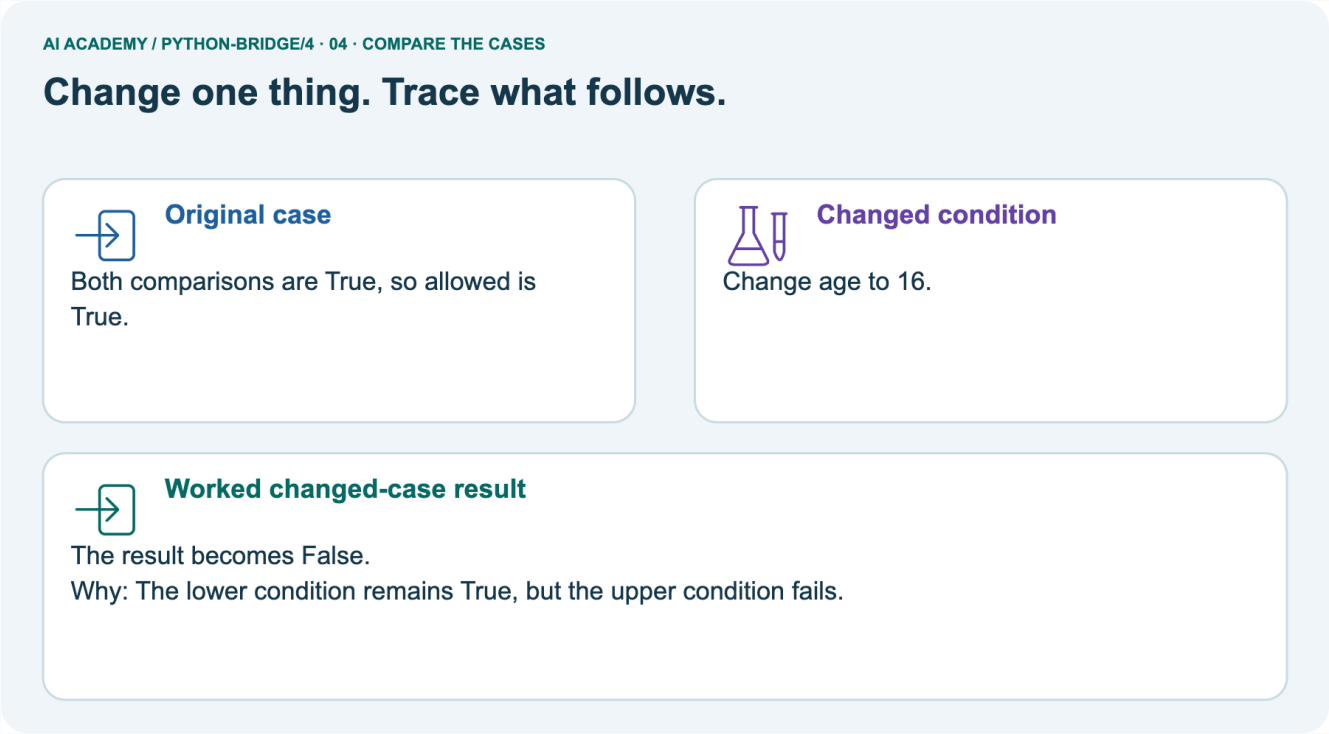


Follow the labelled picture with the matching explanation. Open the vector diagram

Two requirements must both hold. 12 min, quiet False | 12 <= 12 is True | True and False is False. 12 min, quiet True | 12 <= 12 is True | True and True is True. 15 min, quiet True | 15 <= 12 is False | False and True is False

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

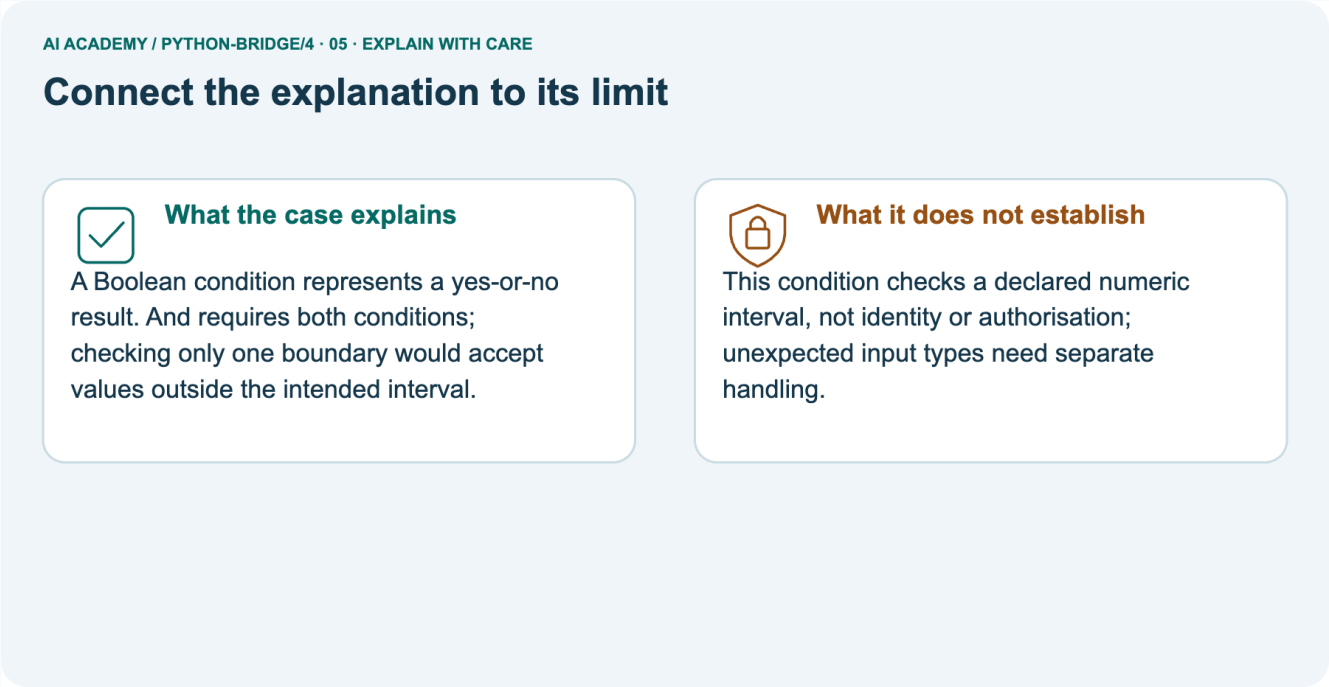


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both **comparison**s are True, so allowed is True. Changed condition: Change age to 16. Expected result: The result becomes False. Reason: The lower condition remains True, but the upper condition fails.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



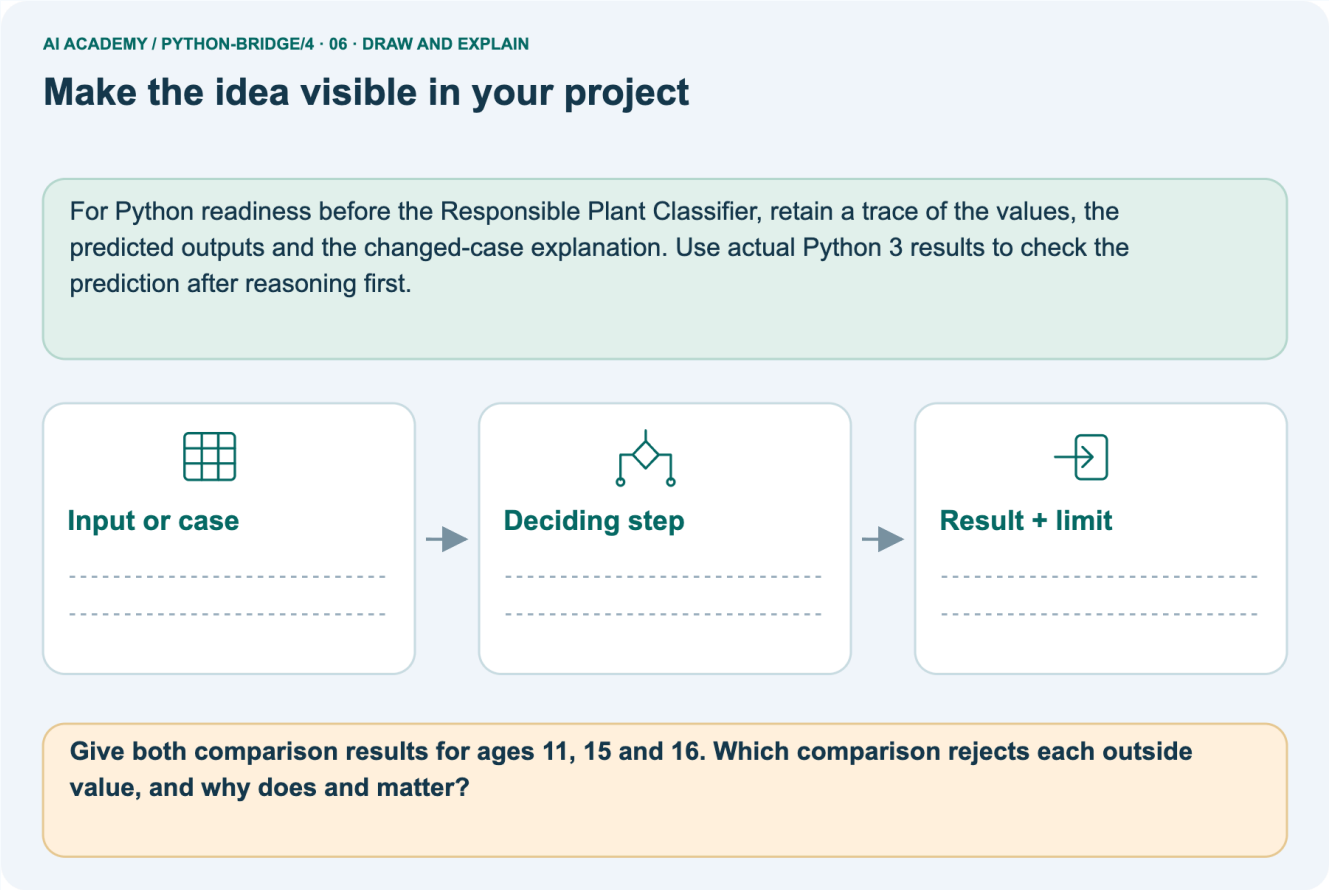
Follow the labelled picture with the matching explanation. Open the vector diagram

A **Boolean** condition represents a yes-or-no result. And requires both conditions; checking only one boundary would accept values outside the intended interval.

Limit: This condition checks a declared numeric interval, not identity or authorisation; unexpected **input** types need separate handling.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Give both **comparison** results for ages 11, 15 and 16. Which **comparison** rejects each outside value, and why does and matter?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Boolean conditions: the case

02. Follow the deciding steps

03. Two requirements must both hold

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.